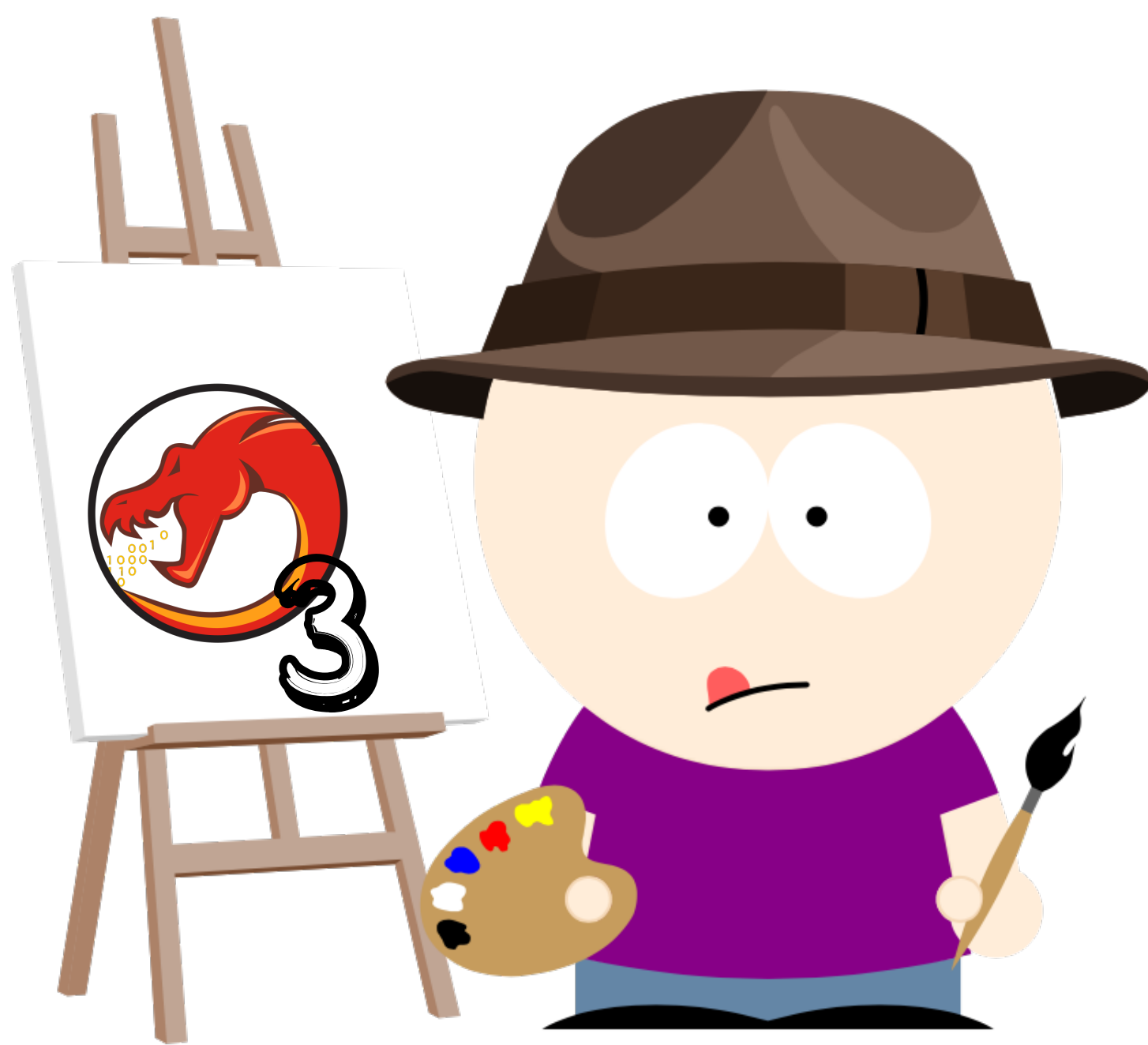




IN-CLASS EXERCISES



Version: 2025-02-08



GHIDRA CODE PATCHING

Download exercises

1. Inside the LigerLabs VM, open up the Firefox browser and go to <https://ligerlabs.org/lectures.html>
2. Download the file
`ghidra-3-exercises.zip`
3. Open up a terminal and Unzip the file
 - > `unzip ghidra-3-exercises.zip`
 - > `cd ghidra-3-exercises`
 - > `ls`

Task 1

1. Load `call1-linux-clang-64.exe`.
2. Currently, `main` calls `foo`.
3. Make `main` call `bar`.
4. Use `VBinDiff` to compare the differences between the original and patched files.

Task 2

1. Load delete1-linux-clang-64.exe:

```
int main (int argc, char** argv) {  
    int t = strncmp(argv[1], "SECRET", 6);  
    if (t != 0) {  
        abort();  
    };  
    puts(roll);  
}
```

2. Delete the call to abort () by overwriting it with NOPs.

Task 3

1. Load `const1-linux-clang-64.exe`.
2. Currently, `main` calls `puts` twice.
3. Make `main` call `puts` 10 times.

Task 4

1. Load insert1-linux-clang-64.exe:

```
char* rick = "Never gonna give you up";  
char* roll = "Never gonna let you down";  
char* you  = "Never gonna run around and desert you";  
  
int main () {  
    puts(rick);  
    asm volatile (".skip 20, 0x90");  
    puts(you);  
}
```

2. Insert a call `puts(roll)` between the two calls to `puts`.

Task 5

1. Load add1-linux-clang-64.exe:

```
#include<stdio.h>

int main () {
    int x = 10;
    x = x + 4;
    printf("%i\n", x);
}
```

2. Change the add instruction to a sub instruction so that the program prints 6 instead of 14.

Task 6

1. Load string1-linux-clang-64.exe:

```
#include<stdio.h>

char* rick = "Never gonna give you up";

int main () {
    puts(rick);
}
```

2. Change the `rick` string so that the program prints
"Always gonna give you up".

Task 7

1. Load `life1-linux-clang-64.exe`:

```
int main (int argc, char** argv) {  
    if (strncmp(argv[1], "hemligt", 7))  
        abort();  
    printf("The meaning of life is %i\n", 42);  
}
```

2. Overwrite the first bytes of `main` with a jump to the `printf`, bypassing the password check.

Task 8

1. Load `life2-linux-clang-64.exe`:

```
int main (int argc, char** argv) {  
    if (strncmp(argv[1], "hemligt", 7))  
        abort();  
    printf("The meaning of life is %i\n", 42);  
}
```

2. Change the branch instruction so that the call to `abort ( )` never executes, regardless of the input.

Task 9

1. Load `cave1-linux-clang-64.exe`. The code is shown on the next page. This is the same example as given in the *Code Cave* section of lecture **ghidra-3**.
2. Follow the same steps as in the lecture to make the program print

Never gonna give you up

Never gonna let you down

Never gonna run around and desert you


```
char* rick = "Never gonna give you up";  
char* roll = "Never gonna let you down";  
char* you = "Never gonna run around and kill no one";
```

```
void codeCave() {  
    L1: asm volatile (".skip 15, 0x90");  
}
```

```
int main () {  
    puts(rick);  
    asm volatile (".skip 5, 0x90");  
    L2: puts(you);  
}
```

Insert
`puts(roll);`
`jmp L2;`

Insert
`jmp L1;`

Task 9 - Steps

1. **Find** code caves

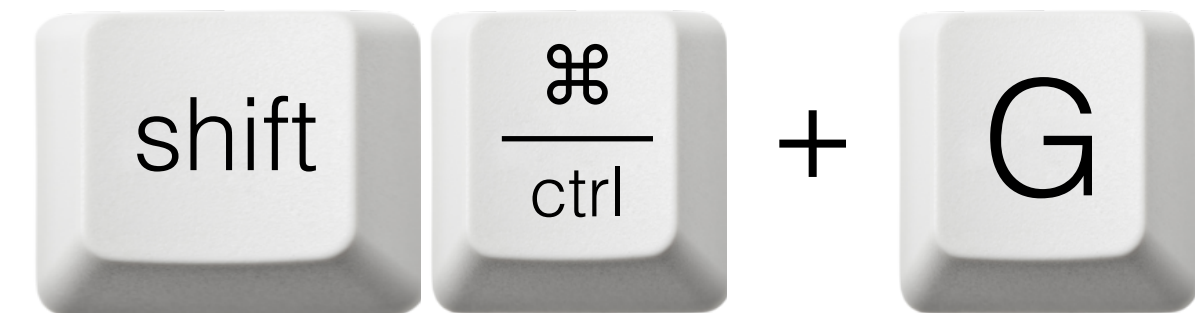
2. **Label** any location you want to refer to



3. **Clear** any bytes you want to modify



4. **Select** the right instruction encoding



5. **Insert**

A) a jump to the code cave

B) the new code you want in the cave

C) a jump back from the cave

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